

Biodegradable Polymer with Hydrolysis-Induced Zwitterions for Antibiofouling

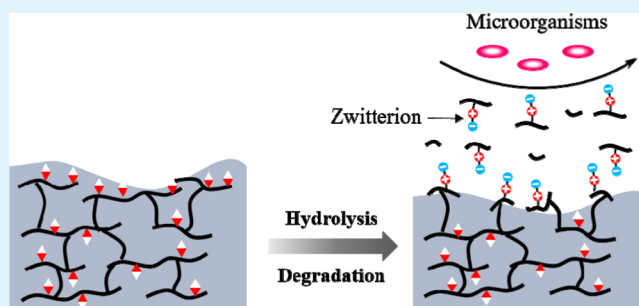
Qingyi Xie,[†] Qianni Xie,[†] Jiansen Pan, Chunfeng Ma,^{*,†} and Guangzhao Zhang^{*,†}

Faculty of Materials Science and Engineering, South China University of Technology, Guangzhou 510640, P. R. China

S Supporting Information

ABSTRACT: Persistent protein resistance is critical for marine antibiofouling. We have prepared copolymer of 2-methylene-1,3-dioxepane (MDO), tertiary carboxybetaine ester (TCB), and 7-methacryloyloxy-4-methylcoumarin (MAMC) via radical ring-opening polymerization, where MDO, TCB, and MAMC make the polymer degradable, protein resistible, and photo-cross-linkable, respectively. Our study shows that the polymer can well adhere to the substrate with controlled degradation and water adsorption rate in artificial seawater (ASW). Particularly, the polymer film can generate zwitterions via surface hydrolysis in ASW. Quartz crystal microbalance with dissipation measurements reveal that such hydrolysis-induced zwitterionic surface can effectively resist nonspecific protein adsorption. Moreover, the surface can inhibit the adhesion of marine bacteria *Pseudomonas* sp. and *Vibrio alginolyticus* as well as clinical bacterium *Escherichia coli*.

KEYWORDS: protein resistance, antifouling, hydrolysis, degradation, radical ring-opening copolymerization



1. INTRODUCTION

It is well known that marine biofouling has a great impact on maritime industries.^{1–3} Because tributyltin-based coatings were banned due to their negative effects on marine environment, much attention was turned to developing eco-friendly antifouling systems, including zwitterionic polymers,^{4–8} silicone-based fouling release coatings,^{9–11} silylated block copolymers,^{12–14} amphiphilic copolymers,^{15,16} and liquid-infused slippery surfaces.^{17,18} Recently, biodegradable polymers were also used in marine antibiofouling in that their degradation owing to seawater and enzyme attack can form a dynamic surface that prevents the organisms or inorganics from landing and adhering even on static conditions.^{19–25} Actually, they exhibit antifouling performance only when the renewal of the dynamic surface or the degradation is quick enough, which reduces their service life. As a compromise, biodegradable polymers are often used with antifoulants for long-term antifouling. Yet, the introduction of antifoulants may have potential impact on the marine environment.

As environmentally friendly systems, zwitterionic polymers can effectively inhibit the protein adsorption and adhesion of microorganisms.²⁶ However, their antifouling performance may decline during their service because they are readily covered by dead bacteria on biomedical conditions or organics and inorganics in marine environment. Namely, zwitterionic polymers usually cannot keep a persistent antifouling ability. Another problem is that zwitterionic polymers are usually swollen in aqueous environments, so their mechanical properties and adhesion to the substrates decrease.²⁷ A combination of zwitterions and biodegradable polymers is expected to

exhibit improved antifouling, mechanical, and environmentally friendly performances. Considering that most biodegradable polymers are synthesized via ring-opening polymerization, it is impossible to prepare such copolymers unless we have cyclic monomers with zwitterionic groups.^{28,29} Radical ring-opening polymerization makes the copolymerization of vinyl zwitterionic monomer and cyclic ketene acetals (CKAs) possible.^{30–33} Yet, it is still a challenge to do that because the polar zwitterions and the hydrophobic CKAs are poorly compatible and they lack co-solvents.³⁴

In the present study, we have prepared novel biodegradable polymer via radical ring-opening polymerization of 2-methylene-1,3-dioxepane (MDO), tertiary carboxybetaine ester (TCB), and 7-methacryloyloxy-4-methylcoumarin (MAMC), where MDO, TCB, and MAMC make the polymer degradable, protein resistible, and photo-cross-linkable, respectively. The degradation of the backbone and hydrolysis of TCB side groups generate a dynamic surface with a controllable erosion rate. Meanwhile, the hydrolysis yields zwitterions with protein resistance and antibacterial property. Particularly, the zwitterions continuously renew with the degradation and hydrolysis so that the surface can possess persistent protein resistance even if the adsorption of organics or inorganics occurs. Because the hydrolysis only occurs on the surface and the generated hydrophilic surface is continuously renewed, the water adsorbed cannot penetrate into the matrix, so that the

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Scheme 1. Synthesis and Hydrolysis of PMCMx

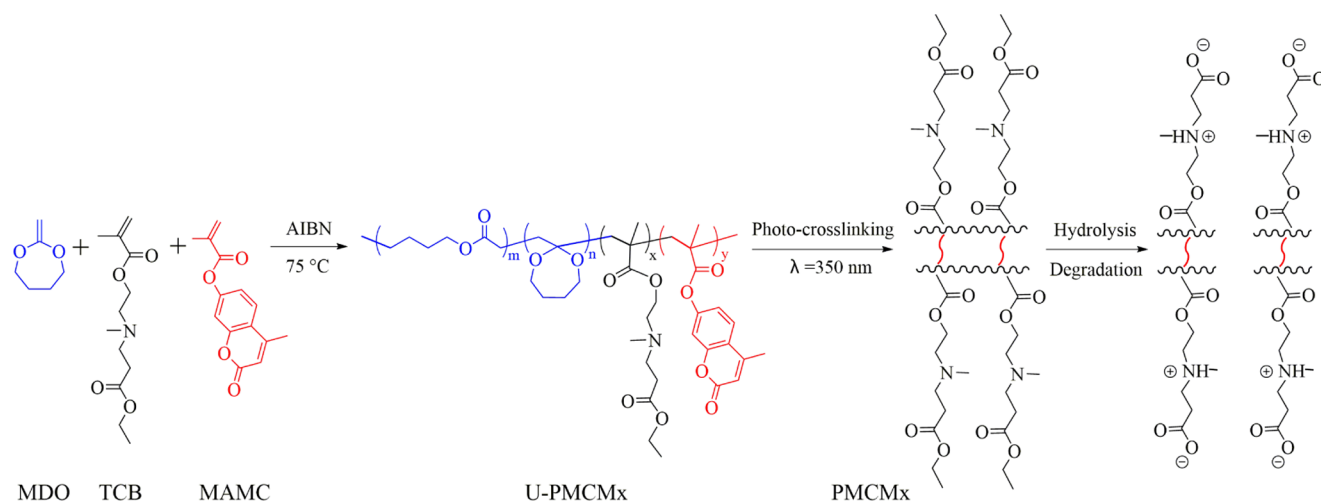


Table 1. Characterization Data of the Polymers

sample	ester/acetal/TCB/MAMC ^a	M_n (kDa) ^b	$\bar{D}_M$ ^b	T_g (°C) ^c	T_g (°C) ^d	γ (MPa) ^e
U-PMCM0	0/0/95.8/4.2	36.3	2.0	-62.3	-51.8	2.28 ± 0.45
U-PMCM9	9.0/2.2/84.3/4.5	13.3	2.1	-40.5	-33.8	2.78 ± 0.20
U-PMCM23	23.0/15.6/57.9/3.5	12.4	3.0	-34.1	-30.8	2.34 ± 0.43
U-PMCM30	30.0/10.8/55.3/3.9	7.4	3.2	-41.0	-32.2	2.24 ± 0.16

^aWeight percentage determined by ¹H NMR. ^bDetermined by gel permeation chromatography (GPC). ^c T_g before photo-cross-linking. ^d T_g after photo-cross-linking. ^eAdhesion strength after photo-cross-linking.

matrix keeps hydrophobic and robust. We have investigated the degradability and antifouling performance and other properties of the polymer with hydrolysis-induced zwitterion (HIZ). Our aim is to develop efficient eco-friendly material for marine antibiofouling.

2. EXPERIMENTAL SECTION

2.1. Materials. 2-Methylene-1,3-dioxepane (MDO), tertiary carboxybetaine ester (TCB), and 7-methacryloyloxy-4-methylcoumarin (MAMC) were synthesized following the procedures described elsewhere.^{35–37} 2,2-Azobisisobutyronitrile (AIBN) was recrystallized twice from methanol. 1,4-Dioxane was refluxed by CaH₂ and distilled prior to use. Artificial seawater (ASW) was prepared following ASTM D1141-98 (2013). Physiological phosphate-buffered saline (PBS, 0.14 M, pH 7.4) was prepared by dissolving NaCl, KCl, Na₂HPO₄, and KH₂PO₄ in Milli-Q water. Lipase from *Pseudomonas cepacia* (PS) was from Aldrich. Fibrinogen (fraction I from human plasma, M_w = 340 kDa, pI 5.5), lysozyme via chicken egg white (M_w = 14.7 kDa, pI 11.1), and bovine serum albumin (BSA) (fraction V from the crystalline form, M_w = 68 kDa, pI 4.8) were purchased from Merck Chemicals.³⁸

2.2. Synthesis of MAMC, TCB, and MDO Copolymer. The synthesis of copolymers consisting of MAMC, TCB, and MDO is shown in Scheme 1. Typically, MDO (1.25 g, 10.96 mmol), TCB (0.7 g, 2.88 mmol), MAMC (0.09 g, 0.37 mmol), AIBN (10.2 mg, 0.5 wt % of total monomers), and 1,4-dioxane (5 mL) were added in a Schlenk tube with a magnetic stirring bar and degassed by three freeze–evacuate–thaw cycles. The polymerization was performed under nitrogen at 75 °C for 48 h. During the copolymerization, 50 μ L of the reaction solution was taken at certain intervals for examining the monomer consumption with reaction time by proton nuclear magnetic resonance spectroscopy (¹H NMR). Finally, the mixture was precipitated into hexane twice and dried under vacuum overnight. The resulting polymer is a yellow viscous paste. For convenience, the copolymer before and after cross-linking are designated as U-PMCMx and PMCMx, respectively, where x is the weight percentage of ester

determined by ¹H NMR. The characterization data is shown in Table 1.

3. CHARACTERIZATION

3.1. Proton Nuclear Magnetic Resonance Spectroscopy (¹H NMR). All ¹H NMR spectra were recorded on a Bruker AV600 NMR spectrometer using CDCl₃ as solvent and tetramethylsilane as the internal standard.

3.2. Gel Permeation Chromatography (GPC). The number-average molecular weight (M_n) and the polydispersity ($\bar{D}_M$) were determined by Agilent gel permeation chromatography (1260 Infinity) with refractive index and UV detectors. The experiment was conducted in tetrahydrofuran (THF) at 35 °C using two identical PLgel columns (5 μ m, MIXED-C) at a flow rate of 1.0 mL/min. Calibration was done with a series of narrowly dispersed polystyrene standards (1.30 $\times$ 10³ to 2.21 $\times$ 10⁶ g/mol).

3.3. Differential Scanning Calorimetry (DSC). DSC measurements were performed on a NETZSCH DSC 204F1 differential scanning calorimeter under a nitrogen flow of 50 mL/min. The sample was quickly heated to 100 °C and equilibrated for 5 min to remove thermal history and then it was cooled to -80 °C at a rate of 5 °C/min. Subsequently, it was heated to 100 °C at a rate of 10 °C/min. The glass transition temperature (T_g) was obtained from the endothermic shift during the second heating scan.

3.4. UV–Vis Spectroscopy. UV–vis spectra were recorded on an Agilent Cary 60 spectrophotometer in the wavelength range of 200–400 nm. U-PMCMx films were prepared by spin-casting from the polymer solution in THF (50.0 mg/mL) on mica wafer. After irradiation by UV light (emitting nominally at 350 nm, light intensity 100%, 100 mW/cm²), the films were measured by the UV–vis spectrophotometer.

3.5. Water Contact Angle (WCA) Measurement. The static water contact angle (WCA) measurements were carried out on a Theta Auto 113 (KSV NIMA) by depositing 3 μL of deionized water droplet on the sample surface using the sessile method. Five different regions from one single sample were measured to obtain an average value. Dynamic WCA was measured by depositing a water droplet of 1 μL on the film surface, increasing its volume to 4 μL and finally decreasing it. The advancing and receding WCA were taken as the maximum and minimum stable angles, respectively.

3.6. Attenuated Total Reflection Infrared (ATR-IR) Spectroscopy. Attenuated total reflection infrared (ATR-IR) spectra were recorded on a Bruker Vertex 70 Fourier transform infrared spectrometer. The spectra were collected at 64 scans with a spectral resolution of 4 cm^{-1} .

3.7. Adhesion Test. The adhesion strength of PMCMx coatings to the glass fiber-reinforced epoxy resin substrate was measured by an adhesion tester (PosiTest AT-A Automatic) according to ASTM Standard D 4541. The films with a thickness of $\sim 300\ \mu\text{m}$ were prepared by casting from polymer solution with subsequent photo-cross-linking. Pull-test adhesion data were obtained by detaching an aluminum dolly (20 mm in diameter) at a speed of 0.2 MPa/s. Five different regions of each sample were tested to obtain an average value.

3.8. Hydrolytic Degradation and Water Adsorption. PMCMx film was prepared by casting from polymer in xylene on an epoxy panel ($20 \times 20\ \text{mm}^2$) and then exposed to UV radiation for 2 h. The weight (W_0) of each dried coating together with its panel was measured before the coating was immersed in ASW. At certain intervals, the sample was taken out and dipped in Milli-Q water for 5 min before its weight ($W_{t,\text{wet}}$) was recorded. Then, the sample was dried before the weight ($W_{t,\text{dry}}$) was recorded. The mass loss was calculated by $(W_0 - W_{t,\text{dry}})/\text{test area}$. The water adsorption was calculated by $(W_{t,\text{wet}} - W_{t,\text{dry}})/\text{test area}$. Each polymer sample was tested in triplicates.

3.9. Quartz Crystal Microbalance with Dissipation (QCM-D). The enzymatic degradation and protein adsorption of PMCMx were investigated by QCM-D from Q-sense AB (Sweden). The films were prepared by spin-casting of PMCMx solution in THF (5.0 mg/mL) on an AT-cut quartz crystal with a fundamental resonance frequency of 5 MHz and then irradiated by an UV lamp for 30 min. The sample was then immersed in ASW for a certain time so that PMCMx was prehydrolyzed. Subsequently, the quartz crystal with PMCMx film was mounted in a fluid cell with one side exposed to the solution. The protein solution (1.0 mg/mL) or lipase PS solution (5 mg/mL) was delivered to the surface at a flow rate of 150 $\mu\text{L}/\text{min}$ using ASW or PBS as reference. Details of the QCM-D measurements can be found elsewhere.³⁹ Briefly, the shift in frequency (Δf) of the crystal is related to the mass change of the thin layer on quartz crystal, whereas the dissipation factor is related to the viscoelastic properties of the additional layer. For a rigid film in vacuum or air, if it is evenly distributed and much thinner than the crystal, the change in frequency (Δf) is related to Δm and the overtone number ($n = 1, 3, 5, \dots$) by the Sauerbrey equation⁴⁰

$$\Delta m = -\frac{\rho_q l_q \Delta f}{f_0 n} = -C \frac{\Delta f}{n}$$

where f_0 is the fundamental frequency, ρ_q and l_q are the specific density and thickness of the quartz crystal, respectively, and C is

the constant of the crystal ($17.7\ \text{ng}/(\text{cm}^2\ \text{Hz})$). The changes in frequency (Δf) and dissipation (ΔD) give information about the protein adsorption or degradation and the structural change of the films. All experiments were performed at 25 $^\circ\text{C}$, and the presented data were from the third overtone ($n = 3$).

3.10. Antibacterial Assays. The antibacterial performance of PMCMx films was evaluated using two types of marine bacteria *Pseudomonas* sp. NOV776 and *Vibrio alginolyticus* ATCC33787 and medically related bacterium *Escherichia coli*. The polymer film was prepared by spin-casting on a glass slide (cut into $1 \times 1\ \text{cm}^2$) with subsequent photo-cross-linking, and it was prehydrolyzed in ASW before using for the antibacterial test. The culture of marine bacteria is follows: the cells of *Pseudomonas* sp. were streaked on MB2216 agar plates (purchased from BD Difco) and grown at 30 $^\circ\text{C}$ for 48 h. A monoclonal colony was inoculated into a tube containing 3.0 mL of MB2216 medium and grown at 150 rpm and 30 $^\circ\text{C}$ for 24 h. The suspension (100 μL) was then reinoculated into 10 mL of MB2216 medium and further cultured for 12 h to mid-log phase. Then, the suspension was centrifuged at 4750 rpm for 5 min, washed by ASW twice, and resuspended in ASW. Finally, the suspension was diluted to $1 \times 10^9\ \text{CFU}/\text{mL}$, determined by a hemocytometer. The cells of *V. alginolyticus* were streaked on MB2216 agar plates and grown at 30 $^\circ\text{C}$ for 24 h. A monoclonal colony was inoculated into a tube containing 3.0 mL of MB2216 medium and grown at 150 rpm and 30 $^\circ\text{C}$ overnight. The culture (100 μL) was then reinoculated into 10 mL of MB2216 medium and further cultured for 3 h to mid-log phase. Then, the suspension was centrifuged at 4750 rpm for 5 min, washed by ASW twice, resuspended in ASW, and finally diluted to $1 \times 10^9\ \text{CFU}/\text{mL}$. For *E. coli*, bacterial cells were streaked on lysogeny broth agar plates (1% tryptone, 0.5% NaCl, 0.5% yeast extract, and 1.5% agar in Millipore-Q water; tryptone yeast extract) and grown at 37 $^\circ\text{C}$ for 24 h. A monoclonal colony was inoculated into a tube containing 3.0 mL of fresh tryptone medium (1% tryptone and 0.5% NaCl in water) and grown at 200 rpm and 33 $^\circ\text{C}$ for 12 h. The suspension (100 μL) was then reinoculated into 10 mL of fresh tryptone medium and further cultured for 4 h to mid-log phase. Then, the suspension was centrifuged at 4750 rpm for 5 min, washed by PBS twice, resuspended in PBS, and finally diluted to $1 \times 10^9\ \text{CFU}/\text{mL}$. Each polymer film was immersed in bacterial suspension under static conditions at 30 $^\circ\text{C}$ for 5 h, washed gently with ASW (for *Pseudomonas* sp. and *V. alginolyticus* assay) or PBS (for *E. coli* assay), and stained with a LIVE/DEAD BacLight Bacterial Viability Kit. The adhered bacterial cells were observed using a fluorescence microscope (Scope A1, Zeiss).

4. RESULTS AND DISCUSSION

The successful copolymerization of U-PMCMx polymers was confirmed by ^1H NMR (Figure S1). The time dependence of monomer consumption during polymerization was also examined (Figure S2). TCB and MAMC copolymerize faster than MDO, which causes a deviation in composition of the copolymer relative to the theoretical prediction. Meanwhile, the copolymerization yields a small proportion of acetal rings besides esters (Figure S1 and Table 1), which is probably due to the large steric hindrance of MAMC.⁴¹ Each copolymer exhibits a single glass transition temperature (T_g) (Figure S4), indicating that all U-PMCMx samples are random copolymers. Moreover, no melting peak can be observed in DSC curves showing that the copolymers are amorphous. The polymer with

such a low T_g is a viscous paste and cannot serve as coating without cross-linking. Thus, photo-cross-linking of U-PMCMx was performed to improve its mechanical property, which was confirmed by UV-vis spectra (Figure S5). T_g of each copolymer increases after cross-linking, which (-52 to -30 °C) is close to the theoretical value calculated from Fox equation with those of MDO homopolymer (-59 °C) and cross-linked TCB (-21 °C).^{30,42} After photo-cross-linking, the viscous paste turns into a solid film. As shown in Table 1, all of the samples have adhesion strength above 2.0 MPa, indicating that the mechanical property of U-PMCMx is improved by photo-cross-linking.

It is well known in organic chemistry that ester groups would hydrolyze into carboxyls in aqueous solution and either alkaline or acid can act as catalyst and speed up the reaction. Accordingly, tertiary carboxybetaine ester (TCB) groups can be hydrolyzed into zwitterions in aqueous solution.^{43,44} We examined the hydrolysis of TCB groups in PMCM9 in ASW by ATR-IR with that in NaOH solution as the control (Figure 1).

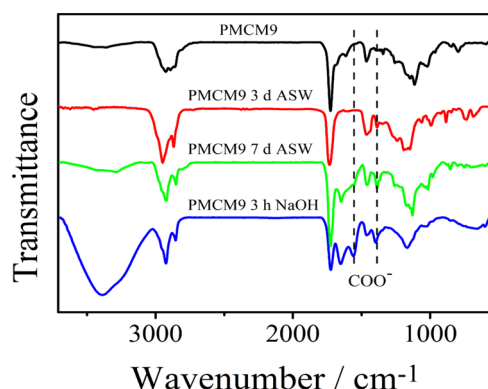


Figure 1. ATR-IR spectra of PMCM9 with different immersion times in ASW and 0.1 M NaOH solution for 3 h.

For PMCM9 immersed in 0.1 M NaOH, new bands at 1398 and 1561 cm^{-1} assigned to carboxyls can be observed, confirming that TCB is hydrolyzed into carboxybetaine groups. The bands at 3387 and 1650 cm^{-1} are attributed to hydroxyl from hydrated zwitterions. The intensity of the bands for carboxyls gradually increases in the spectra of PMCM9 with immersion time in ASW, indicating the formation of zwitterionic surface. Note that the intensity is lower than that in 0.1 M NaOH solution because the alkalinity of ASW is much weaker yielding less zwitterions.

Figure 2 shows the static water contact angles (WCA) of PMCM9 with different immersion times in ASW. The static WCA gradually decreases from about 90 – 80 ° as time increases, indicating the generation of hydrophilic zwitterionic groups on the surface. The receding WCA decreases from ~ 70 to 50 ° (Figure S7), further indicating the presence of zwitterionic group because it mainly depends on the hydrophilic phases on the surface.³ In contrast, WCA of the sample after immersion in 0.1 M NaOH for 3 h decreases to lower than 40 ° because TCB hydrolyzes more rapidly there yielding more zwitterions. Similar results regarding dynamic and static WCA were obtained for all of the copolymers (Figures S7 and S8). This is understandable because TCB content (>50 wt %) in each sample is high enough to cover the surface, leading to a similar wettability.

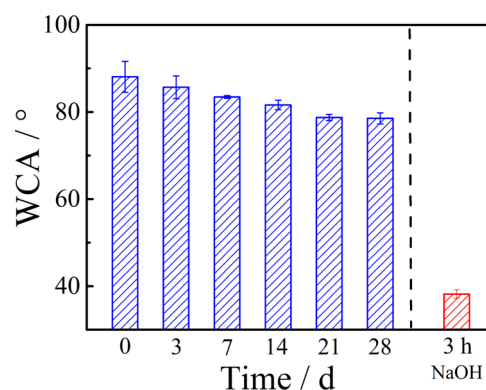


Figure 2. Static water contact angle (WCA) of PMCM9 with different immersion times in ASW (blue) at 25 °C with that immersed in 0.1 M NaOH solution for 3 h as the control (red).

Figure 3 shows the adhesion strength of PMCM9 with different immersion times in ASW. The adhesion strength

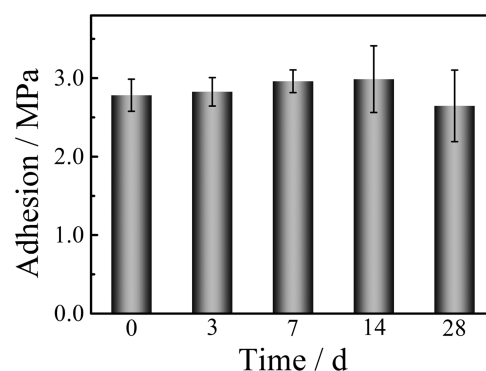


Figure 3. Adhesion strength of PMCM9 with different immersion times in ASW at 25 °C.

slightly varies with time, but it is always above 2.0 MPa, indicating that PMCM9 has good adhesion to substrates. This is because TCB hydrolysis only happens on the surface and the film is relatively hydrophobic, as revealed by WCA test. Moreover, the degradable main chains facilitate the removal of hydrophilic surface. Thus, the matrix remains hydrophobic and slightly swells.

Under the attack of enzyme excreted by microorganisms, biodegradable polymers can degrade in marine environment yielding dynamic surface that can prevent the settlement of fouling organism.⁴⁵ The enzymatic degradation of PMCMx was evaluated using QCM-D. Figure 4 shows the time dependence of frequency shift (Δf) and dissipation shift (ΔD) regarding the enzymatic degradation of the PMCMx in ASW. For PMCM0, the decrease in Δf indicates the adsorption of lipase PS on PMCM0 film. For PMCM9, Δf decreases relative to the baseline but is smaller than that of PMCM0. This is because the degradation and adsorption of lipase PS lead to increase and decrease in frequency at the same time, respectively, and the latter overcomes the former. For PMCM23 and PMCM30 with higher MDO contents, Δf exhibits notable increase relative to the baseline after rinsing, indicating the mass loss of the films. Namely, the main chain degrades into small fragments, which subsequently disperse into the solution. Clearly, the degradation rate increases with MDO content, as reflected by higher increase in Δf . On the other hand, the increase in ΔD for

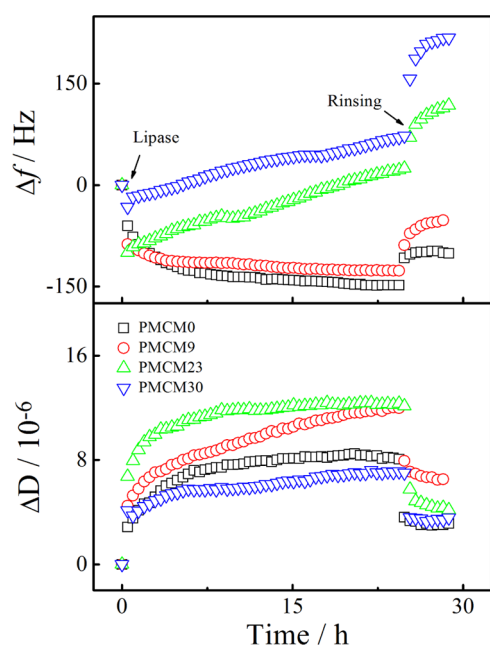


Figure 4. Time dependence of the frequency shift (Δf) and dissipation shift (ΔD) regarding the enzymatic degradation of PMCMx films in ASW at 25 °C.

PMCM0 demonstrates the formation of a viscoelastic layer of lipase PS. For PMCM9 and PMCM23, the increase in ΔD is higher than that for PMCM0. This is because their degradation leads to a surface with a nonuniform structure.⁴⁶ PMCM30 with the highest MDO content shows a relatively less increase in ΔD , indicating a lower thickness resulted from fast degradation. It is well recognized that degradation is affected by the content of cleavable linkage, wettability, cross-link density, and crystallinity.⁶ Considering that all these amorphous PMCMx samples have a similar cross-link density (Table 1), the biodegradation rate is regulated by MDO content. Note that the average molar mass (M_c) between two neighboring cross-links can influence the biodegradation rate in principle. Considering that the samples have similar cross-linking density, they have similar M_c , so its effect can be neglected.

We examined the hydrolytic degradation of PMCMx. Figure 5a shows the time dependence of mass loss of PMCMx after immersion in ASW at 25 °C. After 3 days, the mass of all PMCMx films starts to decrease. For PMCM0, its mass loss is attributed to the hydrolysis of TCB and the consequent dissolution of hydrophilic zwitterionic fragments. PMCM9

loses weight faster than PMCM0 because the cleavage of the ester linkages in the backbone can assist the hydrolyzed zwitterionic segments to dissolve into ASW. However, further increasing MDO reduces the mass loss rate (PMCM23 and PMCM30). This is because TCB content decreases with MDO content slowing down the hydrolysis. At similar TCB contents (57.9 and 55.3 wt %), PMCM30 has a mass loss rate higher than that of PMCM23 because more MDO leads to faster degradation of main chains. Clearly, the mass loss rate is determined by both degradation of main chains and the hydrolysis of side groups. An optimum self-renewal rate can be achieved by varying polymer composition. Figure 5b shows the time dependence of water adsorption of PMCMx after immersion in ASW at 25 °C. The water adsorption of PMCMx increases as MDO content decreases or TCB content increases. This is understandable because a higher TCB content leads to more hydrophilic zwitterionic groups and higher hydration. In other words, the incorporation of MDO into copolymer backbone can effectively reduce the swelling of polymer, which is favorable to its applications in aqueous environment.

The protein resistance on PMCM9 film without and with prehydrolysis in ASW was evaluated by QCM-D using fibrinogen, bovine serum albumin (BSA), and lysozyme. Figure 6 shows the time dependence of frequency shift (Δf) and dissipation shift (ΔD) of the adsorption of fibrinogen, which is a large plasma protein carrying negative charge in ASW. With regard to the PMCM9 without prehydrolysis, as fibrinogen is introduced, Δf decreases and gradually levels off. After rinsing with ASW, Δf still exhibits a notable decrease relative to the baseline. It is known that the frequency (Δf) decreases with the increasing mass or thickness of the layer, whereas the dissipation (ΔD) increases with thickness but decreases as the rigidity increases. In other words, the decrease in Δf and increase in ΔD indicate that the fibrinogen adsorbs on the PMCM9 with prehydrolysis and forms a viscoelastic layer. With prehydrolysis in ASW for 3 days, the changes in Δf and ΔD become much smaller, indicating less protein is adsorbed. As the prehydrolysis time increases to 14 days, Δf and ΔD after rinsing, return to baseline, indicating no adsorption of fibrinogen.

The adsorption of BSA and lysozyme was also evaluated by QCM-D. The final frequency shifts (Δf) about the three proteins are summarized in Figure 7. Clearly, the adsorption of all of the proteins decreases with the prehydrolysis time because TCB is hydrolyzed into zwitterionic groups in ASW forming a protein-resistant surface. Actually, the zwitterions are

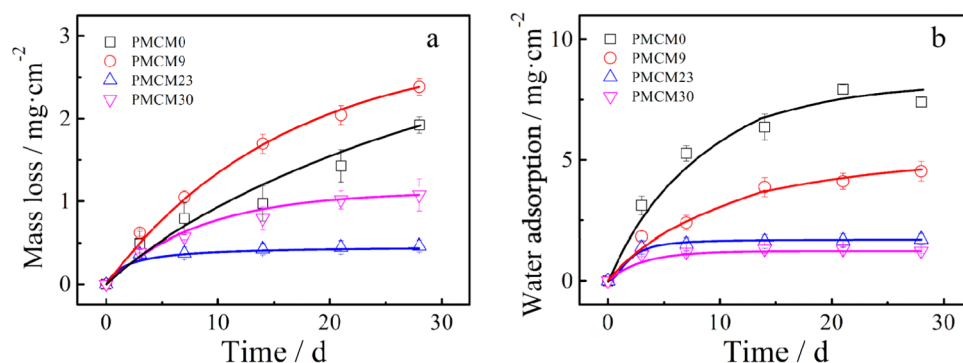


Figure 5. Time dependence of mass loss (a) and water adsorption (b) of PMCMx in ASW at 25 °C.

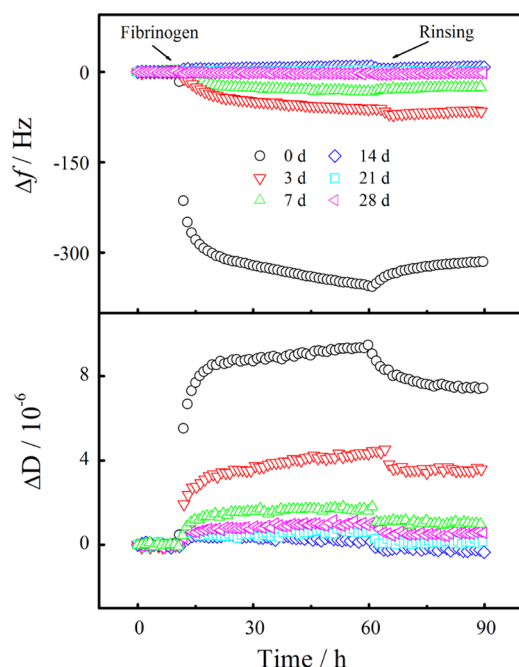


Figure 6. Frequency shift (Δf) and dissipation shift (ΔD) regarding the adsorption of fibrinogen on PMCM9 immersed in ASW with different prehydrolysis times in ASW at 25 °C.

continuously generated by the hydrolysis, whereas the surface is renewed by the degradation of main chain and hydrolysis of the side groups, so that zwitterions on the surface keep active and the surface has a persistent protein resistance.

As we know, biodegradable polymers have been used in drug delivery, implantable devices, and tissue engineering, where they often suffer from protein adsorption or bacterial adhesion, which cause medical contamination and infection.^{47,48} To explore possible application of the biodegradable polymers in medical field, we also examined the adsorption of the proteins on PMCM9 surfaces with prehydrolysis in PBS. The surfaces can effectively resist the nonspecific protein adsorption (Figure S9), indicating that they might be used in medical environment.

We evaluated the antibacterial efficiency of the polymer surfaces. Figure 8 shows the fluorescence microscopy images of PMCM9 surfaces after exposure to ASW suspension of marine bacteria *Pseudomonas* sp. and *V. alginolyticus* for 5 h. Here, we used poly(ϵ -caprolactone) (PCL) without TCB as negative control and PMCM9 with highly dense zwitterions prepared by prehydrolysis in 0.1 M NaOH solution as positive control. A large crowd of both types of bacterial cells adhered on PCL

surface. In contrast, no bacteria are observed at PMCM9 surface with prehydrolysis in 0.1 M NaOH solution for 3 h, indicating that zwitterionic groups generated by TCB hydrolysis can effectively resist the adhesion of bacteria. For PMCM9 with prehydrolysis in ASW for 3 days, almost no bacteria are adhered on the surface. This is understandable because PMCM9 already generated some zwitterions. PMCM9 with prehydrolysis in PBS for 3 days also shows excellent antibacterial efficiency toward *E. coli* (Figure S10). The facts further indicate that the biodegradable polymer PMCMx with hydrolysis-induced zwitterions can be used in medical environment. Note that no bacterial adhesion occurs even when PMCM9 is exposed to bacterial suspension for 5 h without prehydrolysis in ASW or PBS (Figure 8), indicating that PMCM9 would hydrolyze and generate enough zwitterions once it contacts with aqueous solution owing to the high TCB content (84.3 wt %). We also used *Pseudomonas* sp. to examine the effect of TCB or MDO content on the antibacterial ability (Figure S11). We did not observe bacterial adhesion on PMCM0 surface containing 95.8 wt % of TCB either with or without prehydrolysis. On the contrary, a lot of bacteria are adhered on PMCM23 and PMCM30 without prehydrolysis because their TCB contents (57.9 wt % for PMCM23 and 55.3 wt % for PMCM30) are not high enough to generate sufficient zwitterions during the 5 h exposure to bacteria suspension. Clearly, the antibacterial ability of PMCMx without prehydrolysis increases with TCB content. Nevertheless, all of the copolymers can effectively inhibit the bacterial adhesion after prehydrolysis in ASW for 3 days, indicating their excellent antibacterial performance.

5. CONCLUSIONS

In conclusion, we have prepared biodegradable polymer with hydrolysis-induced zwitterions via radical ring-opening polymerization of 2-methylene-1,3-dioxepane (MDO), tertiary carboxybetaine ester (TCB), and 7-methacryloyloxy-4-methylcoumarin (MAMC). The degradation of the main chain and hydrolysis of side groups lead to a dynamic surface that continuously renews. The hydrolysis of side groups yields zwitterions on the surface. The hydrolysis-induced zwitterion (HIZ) surface exhibits persistent protein resistance and antibacterial performance. The photo-cross-linking allows the polymer to form a network with good mechanical strength. The polymer with relative hydrophobicity provides good adhesion to substrates. Such polymer is expected to find marine antibiofouling and biomedical applications.

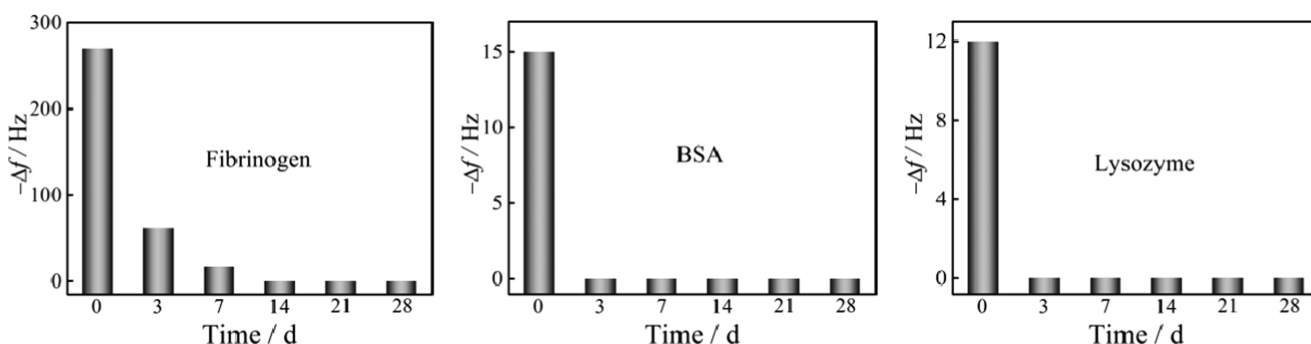


Figure 7. Frequency shift (Δf) regarding the adsorption of different proteins on PMCM9 with different prehydrolysis times at 25 °C.

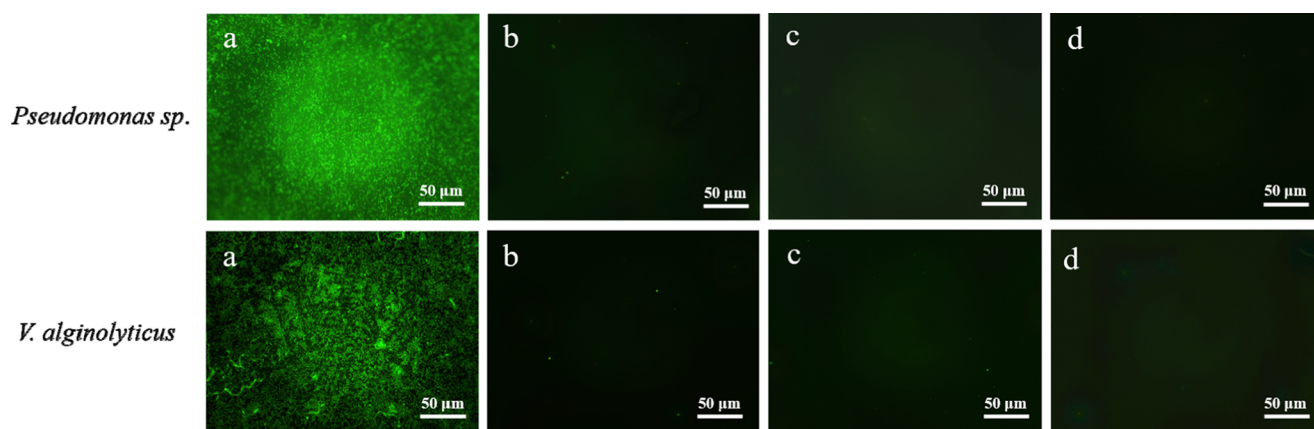


Figure 8. Fluorescence images of *Pseudomonas* sp. and *V. alginolyticus* (ASW suspension) adhered on different surfaces: (a) PCL as negative control, (b) PMCM9 without prehydrolysis, (c) PMCM9 with prehydrolysis in ASW for 3 days, and (d) PMCM9 with prehydrolysis in 0.1 M NaOH solution for 3 h as positive control.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.8b00962.

Time dependence of monomer conversion during polymerization, ^1H NMR spectra, DSC curves of U-PMCM x and PMCM x , GPC curves of U-PMCM x , WCA of PMCM x , UV-vis spectra of U-PMCM9 during photo-cross-linking, QCM-D results of protein adsorption on PMCM9 surfaces in PBS, fluorescence images of *E. coli* on PMCM9 surfaces, and fluorescence images of *Pseudomonas* sp. on PMCM x (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

*E-mail: msmcf@scut.edu.cn (C.M.).

*E-mail: msgzzhang@scut.edu.cn (G.Z.).

ORCID

Chunfeng Ma: 0000-0002-1649-723X

Guangzhao Zhang: 0000-0002-0219-3729

Author Contributions

[†]Q.X. and Q.X. equally contribute to the work.

Notes

The authors declare no competing financial interest.

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